Survey of loss functions for training deep-learning models in computer vision, categorized based on tasks, metric-forms and learning paradigms.

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Abstract

In the era of modern computer vision (CV), deep learning (DL)-based methods have gained significant importance due to their remarkable performance in various applications such as object detection, recognition, segmentation, classification, and restoration in real-world scenarios. Currently, most of the existing survey papers for DL on CV applications provide adequate descriptions of a few methods used for only one or few task(s) in CV, such as object detection, recognition, and segmentation. The success of DL architectures relies profoundly on the effectiveness of loss functions used, appropriate for both the task and DL architecture being employed, especially for CV applications. This paper presents a comprehensive survey of various categories of loss functions used in several computer vision tasks. We have categorized them separately based on the application tasks being addressed, learning paradigms as well as the nature of their metric structure. Three categorized tables containing mathematical expressions of loss functions are cross-linked to extract better knowledge for any sub-group of loss functions. Organization of this survey will benefit researchers/developers to gain knowledge, identify, and use proper cost/energy functions for training DL models with architectural variants, for targeted application(s). Also, an organized list of loss functions, popularly used in the recent past, immensely benefits in the design of novel cost functions for DL. Our motivation and associated contributions stems from the above intentions and issues. Furthermore, we conduct ablation studies with a few widely used effective loss functions applied to address three prominent CV tasks, to illustrate the utility of this survey.

Keywords: Deep Learning (DL), Computer Vision (CV), Survey/Review, Loss/Cost Functions, Categorized Tables, Learning Paradigm, Metric Structure.

1. Introduction

Computer vision refers to the traditional methods and techniques used for processing and analyzing images and videos using handcrafted features and algorithms. Before the advent of deep learning, classical computer vision primarily relied on techniques like edge detection, feature extraction, template matching, and various filters to perform tasks such as object detection, image segmentation, and recognition [1]. To solve many inverse and ill-posed problems in the field of computer Vision, various optimizations functions (with regularization terms) were used to obtain the best possible solution. However, most of these had limitations in obtaining the best desired performance.

Deep learning-based computer vision methods emerged as a major breakthrough in this field with the introduction of convolutional neural networks (CNNs) [2]. The foundation of deep learning dates back to the 1940s and 1950s, but it wasn't until the early 2010s that deep learning gained widespread recognition and started revolutionizing computer vision tasks.

In 2012, a pivotal moment came when a CNN-based architecture called AlexNet [3] won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) [4] by a significant margin, drastically

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reducing the error rate compared to traditional approaches. AlexNet, developed by K. Alex et al. [3], showcased the power of deep learning in handling large-scale visual recognition tasks.

The success of AlexNet triggered a surge of interest in deep learning for computer vision. Researchers and engineers began exploring various CNN architectures and deep learning (DL) techniques for a wide range of computer vision (CV) tasks. Over the years, deep learning models have continued to outperform classical computer vision (CV) methods in various benchmark tasks, including image classification [5, 6, 7, 8], image reconstruction [9, 10], object detection (OD) [11, 12, 13], semantic segmentation [14], image/video generation [15, 16], and several other use cases [17, 18, 19, 20, 21, 22].

Today, deep learning-based computer vision has become the dominant approach due to its ability to automatically learn relevant features and representations directly from the data, making it more adaptable and robust to various real-world scenarios. It has enabled significant advancements in areas like autonomous vehicles, medical imaging, facial recognition, augmented reality, and more, transforming how computers perceive and understand visual information.

Loss functions play a vital role in the success of deep learning models for computer vision tasks. They guide models in understanding visual data by measuring the difference between predictions and actual values. This helps the models to adjust their parameters for better results. A suitable loss function is essential for adequately training a DL model, for any CV tasks like object detection, image segmentation, and restoration. Different tasks need specific loss functions to capture data traits. So, grasping various loss functions is key to creating successful computer vision models.

In computer vision tasks, several types of loss functions are commonly employed to train and optimize deep learning models. Some of the prominent ones include:

- Mean Squared Error (MSE): Often used for regression tasks, it calculates the average of the squared differences between predicted and actual values.
- Binary Cross-Entropy (BCE): Utilized for binary classification problems, it measures the dissimilarity between predicted probabilities and ground truth binary labels.
- Categorical Cross-Entropy (CCE): Applied to multiclass classification tasks, it quantifies the difference between predicted class probabilities and true class labels.
- Dice Loss: Primarily used for image segmentation, it gauges the overlap between predicted and actual segmentation masks.
- Focal Loss: Useful for addressing the class imbalance in object detection, it assigns different weights to well-classified and hard-to-classified samples.
- Wasserstein Loss: Employed in generative adversarial networks (GANs) to measure the distance between real and generated data distributions.
- Perceptual Loss: Used for image style transfer and super-resolution tasks, it measures the difference in feature maps extracted by pre-trained convolutional neural networks.
- Adversarial Loss: Commonly seen in GANs, it encourages the generator to create data that is indistinguishable from real data by minimizing the difference between real and generated distributions.
- KLD-Loss: KL Divergence Loss, or Kullback-Leibler divergence, quantifies the difference between two probability distributions by measuring the information lost when approximating one with the other.

These loss functions address different aspects of computer vision tasks and help DL models learn and improve over training iterations. The choice of the appropriate loss function depends on the specific task and the desired characteristics of the model's output. A supplementary document here associated with this manuscript, details most of the loss functions referred in this manuscript.

One of the methods commonly used by CV (with DL) researchers is to typically perform an ablation study using few DL architectures combined with several loss function components, to obtain the best performing combination(s) to solve a task. To facilitate this work comfortably, one should possess a list (table) of several loss functions commonly (or even rarely) used for training on the DL architecture, as available in published literature, for solutions to the task at hand. This has been our key motivation behind formulating (devising) this article and hence framing its layout (with apt contents) along these specific needs. Note that the focus of this survey paper is on various loss functions used to solve problems in CV (with DL). There are many other survey papers published and thus can be referred - on DL architectures used [23, 24], datasets [25, 26] used for training and performance of several methods [27, 28]; all of which we have not dealt within this article to avoid repetition (duplication).

In this paper, we provide tables for different categories of loss functions used, based on : (i) the Application domain; (ii) the metric nature of formulation, and (iii) the Learning paradigm of the DL Model. Section 2 provides a brief review of a selected set of survey papers published in the literature on DL algorithms for CV tasks. At the end of section 2 we highlights few short coming (with respect to our goals) in those survey papers, and mention how they are addressed by us. Section 3 provides the tables of different categories of loss function, which are all discussed in a supplementary doc. Section 4 provides ablations studies for certain tasks such as classification and recognition/detection. Section 5 concludes the paper with comments. Kindly note that, we do not discriminate between the terms "review" and "survey", and may often use these terms interchangeably (inspite of their subtle variations in meaning, context and applicability).

2. Brief highlights of survey papers published on DL for CV tasks

Various review and survey papers [29], [30], [31], [32], [33], [34], [35], [36], [37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47] [48, 49, 50, 51, 52, 53, 36, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66] have been published on deep learning-based methods focusing on applications like healthcare, deployment platforms, and data repositories, deep neural network architectures, self-supervision learning in Graph Neural Networks (GNNs) with a special emphasis on contrastive and predictive models, facial expression analysis, and image segmentation. Each paper contributes to the understanding and exploration of deep learning algorithms in their specific application domain, providing valuable insights into their usage, performance, and implications. Brief highlights for few have been provided below. We discuss a few prominent ones only, as available in literature. At the end of this section, we identified the gaps and highlight our intent and the contents provided in our survey paper.

The survey paper by Monnier et. al [67] explores solutions to improve the quality and accuracy of video segmentation (VS) task, in addition to that for improving the efficiency in terms of inference time. Datasets related to the semantic video segmentation task and the challenges involved are also discussed. The goal of this survey was not to provide an exhaustive list of all video segmentation approaches, rather to give some useful hints and form a basis on which one can build upon in future work towards practical and efficient video segmentation methods. After providing a background of various approaches used of VS, the authors present ideas of DL (and related ML) architectures used for the task such as – 3D convolution, Attention, optical flow, recurrent, CRF, query-based architectures etc. A short sub-section discusses loss functions, but no formulation/expression is given and discussed (appears to be not in the main focus of the paper). Major section of the paper discusses methods of accelerating the inference, using multi-scale, early-layer features, reducing redundancy, kernel factorization, fast transformers, pruning etc. No tables provided (except one for datasets) – few diagrams with architectures illustrated in the paper.

Abdel-Jaber et. al [29] explores the application of deep learning algorithms in healthcare, focusing on their accuracy. Whereas, [30] concentrates on the collection of popular deep learning deployment platforms, categorizing surveyed articles by broad application areas. The paper also delves into data repositories for deep learning practitioners, offering a comprehensive overview of the distribution of resources within the field.

Chang et. al in [39] first summarize the general definition of the scene graph, and then conduct a comprehensive and systematic discussion on the generation method of the scene graph. They also investigated the main applications of scene graphs and summarized the most commonly used datasets.

Introducing the four pillars for small object detection, including multiscale representation, contextual information, super-resolution, and region-proposal, the researchers in the article [40] investigated the state-of-the-art small object detection networks along with a special focus on the differences and modifications to improve the detection performance comparing to generic object detection architectures.

Recently transformer-based models [68] perform similar to or better than other types of networks such as convolutional and recurrent neural networks. Given its high performance and less need for vision-specific inductive bias, transformers have received better attention from the computer vision community. Han et. al in [41] reviews these vision transformer models by categorizing them in different tasks and analyzing their advantages and disadvantages.

Shrestha and Mahmood in [31] delves into Deep Learning algorithms, providing a tutorial-style overview with comparison tables of frameworks, libraries, and network architectures for maritime object recognition (OR). It explores fundamental Deep Learning algorithms and architectures, presenting a comprehensive guide.

Wang et. al [32] provides an extensive overview of popular deep learning frameworks and libraries, presenting a detailed comparison table for deep neural network (DNN) architectures. The table encompasses architectural details, training specifics, frameworks used, and datasets employed, facilitating a thorough understanding of the landscape of deep learning algorithms.

Xie et. al [33] introduces a figure in a tree-type structure, illustrating self-supervision learning in Graph Neural Networks (GNNs). The paper summarizes different self-supervised learning settings and the corresponding datasets used in each setting, with a special focus on contrastive and predictive models applied to specific applications.

Brugnar et. al [38] focuses on defining a comprehensive taxonomy, comparing different approaches to propose an updated and general framework of hand-based methods in first-person vision (FPV), highlighting the current trends and summarizing the main findings, to researchers who want to improve and expand this field of research with available datasets in this context.

Ben et. al [35] delves into facial expression analysis, outlining the main differences between macro- and micro-expressions. The paper provides details on publicly released micro-expression datasets, introduces the Emotional Facial Action Coding System for macro-expressions, and evaluates recognition algorithms, offering insights into facial expression recognition.

Minaee et. al [36] categorizes various models for image segmentation, providing descriptions of fully convolutional networks, convolutional models with graphical models, encoder-decoder-based models, and others. While the paper includes numerous architectural diagrams, it emphasizes the importance of these models in the context of image segmentation applications, highlighting their characteristics.

A review paper by Fatemeh Shafizadegan et al. [46] categorizes existing methods of Vision-based Human Action Recognition (HAR) only and categorizes them into four levels, providing an in-depth and comparable analysis of approaches – mainly comparing the performance of state-of-the-art methods on benchmark datasets. The paper initially presents a table for Recent surveys on HAR, where only the main focus is mentioned. The 2nd table provided different visual modalities used in HAR, for a benchmark dataset. A 3rd table provides a list of benchmark Multimodal vision-based action datasets. The last table provides a list of methods with the best accuracy.

The paper by Hao Liu et al. [69] presents a systematic review of the current developments of incremental learning and give the overall taxonomy of the incremental learning methods. Specifically, three kinds of mainstream methods, i.e., parameter regularization based approaches, knowledge distillation-based approaches, and dynamic architecture-based approaches, are surveyed, summarized and discussed. The performance of data-permuted incremental learning, class incremental learning, and multi-modal incremental learning on commonly used datasets are comprehensively analysed, for image classification and semantic segmentation.

Yang et. al [34] lacks a table or categorization diagram but addresses the development of effective optimization objectives for deep single-image super-resolution (SISR) learning. The paper establishes a baseline and identifies critical limitations, contributing to the understanding of deep learning for super-resolution, especially in the context of its application.

A survey paper [42] by Geng et. al provides a comprehensive survey of existing open set recognition techniques covering representations of models, datasets, evaluation criteria, and comparing algorithms. The limitations of existing approaches are highlighted. Most of the discussion is on traditional ML methods and few generative models for OSR (Open Set Recognition). A subsection discusses DL-based methods, without any loss functions or training paradigms being mentioned. Many tables provided discuss categories of OSR, comparison of software packages, performance accuracies etc.

The paper by Yiwen Xu et al. [43] presents an overview of the foundational principles and technical advantages of edge deep learning. It also highlights a novel categorization of edge hardware platforms based on performance and usage scenarios, facilitating platform selection and operational effectiveness, focusing on lightweight design and model compression. Focus is on applications, without any discussions on learning paradigms and cost functions.

The paper by Xia Zhao et al. [44] presents an elementary understanding of CNN components and their functions, including input-output, pooling, fully connected & convolution layers; activation functions, batch normalization, dropout. The authors provide a comprehensive overview of the past and current research status of the applications of CNN models in computer vision fields, e.g., image classification, object detection, and video prediction. Treatment focuses on soft-activation functions, attention, dropout, with limited discussion on cost functions and learning strategies.

The survey paper by Asifullah Khan et al. [45] presents a taxonomy of the recent vision transformer architectures and more specifically that of the hybrid vision transformers (ViT). Focus is on key features such as the attention mechanisms, positional embeddings, multi-scale processing, and convolution. This survey stresses on the emerging trend of hybrid vision transformers, across a range of computer vision tasks. The three tables provided are for : (i) online resources relevant to ViT; comparison of strengths, weaknesses, rationale, and performance of several (ii) ViT and (iii) HVT (Hybrid vision transformer) architectures on benchmark datasets. No analysis of cost functions and training paradigms are provided.

The paper by C Meyers et al. [70] presents and exhaustive empirical survey of attacks and defenses and present a simple and practical framework for analyzing any machine-learning system from a safety-critical perspective using adversarial noise to find the upper bound of the failure rate. They conclude that all tested configurations of the ResNet architecture fail to meet any reasonable definition of 'safety-critical' when tested on even small-scale benchmark data. Examining state of the art defenses and attacks against computer vision systems with a focus on safety-critical applications in autonomous driving, industrial control, and healthcare, they conclude that modern neural networks consistently fail to meet established safety-critical standards by a wide margin. The paper presents only one table with failure rates. Analytics are barely presented on attack functions and not on cost functions for optimization.

The survey paper by Wang & Zhu [71], integration of different context information in computer vision tasks is reviewed. Context is categorized into different types (levels), in addition to discussion on machine learning models and image/video datasets that can employ context information. Context-based integration and context-free integration over two classes of tasks: image-based and video-based, are compared. Tables are provided for - summary of datasets used; list of methods for Image and video-based context integration; another one gives for merits of the same. No loss function or training paradigms are discussed – perhaps not in focus of the paper. Lot of figures discuss results and few prominent architectures.

The survey paper by Alfarano et. al [72] provides an overview of optical flow techniques and their application. For a comprehensive review, this survey covers both classical frameworks and the latest AI-based techniques. The limitations of current benchmarks and metrics are highlighted, underscoring the need for more representative datasets and comprehensive evaluation methods. The survey also highlights the importance of integrating industry knowledge and adopting training practices optimized for deep learning-based models. Two tables provided – one gives comparison of performance, and the other one on datasets used. Challenges faced and results are also discussed. The paper concludes that current benchmarks and measurements limit further progress and do not adequately represent real-world scenarios. Preliminary equations for optical flow are discussed, but no loss functions for training DL methods are given.

The survey paper by Dibenedetto et. al [73] presents a comprehensive summary of the latest developments in DL-based 2D and 3D HPE (Human Pose Estimation) approaches over various application domains. Different techniques, datasets (RGB monocular and multi-view images and videos), and evaluation metrics utilized in this field are explored, where the significant advancements and unresolved issues requiring attention to enhance the precision, resilience, and utility of HPE algorithms are highlighted. The paper presents classification of the approaches, use-case examples, and benchmarks over most advanced outcomes in the HPE research field. The paper starts with background and preliminaries, then discusses various approaches and techniques, followed by frameworks-tools-libraries, datasets and evaluation metrics and lastly on challenges to be faced in this area of work. Tables are provided for summarizing and comparing prominent 2D HPE approaches, datasets used (separately for 2D and 3D), comparing accuracies etc. Not a single loss function (except the metrics for performance evaluation) or architectural diagrams (named though) or training paradigm given or discussed in detail.

The paper by Chen et. al [74] also summarizes several approaches for HPE – 2D, 3D of single person and multiple persons using tables; backbone and highlights are also briefed. No discussion of loss functions provided. Datasets used are elaborated.

The survey paper by Velastegui et. al [75] focuses on comprehensive understanding of the strengths and weaknesses of semantic segmentation methods in indoor scenarios to enhance the applicability. The evaluation also consists of a performance analysis of the methods in prevalent real-world segmentation scenarios that pose challenges. The key contributions are: 50 DL architectures for segmentation; evaluation in terms of speed, temporal consistency and vulnerability to corruption; performance analysis over noise, blur, optical aberrations etc. Tables are provided for short survey of 8 papers; comparison of segmentation performance in terms of accuracy, temporal consistency, and inference speed; and for average corruption vulnerability. Loss functions are not in focus, except those evaluation metrics – IoU, Dice etc.

The paper by Simone F et. al [76] provides an interesting survey on bias in visual datasets. Categorization of the bias is done too. A key conclusion of their study is that the problem of bias discovery and quantification in visual datasets is still open, and there is room for improvement in terms of both methods and the range of biases that can be addressed.

The paper by J Wang et. al [77] presents a thorough review of existing deep learning based works for 3D pose estimation and summarizes the advantages/disadvantages of these methods. Furthermore, a comprehensive study for comparison and analysis are conducted on the commonly used benchmark datasets. According to the kinds of models used, the representations of poses to skeleton and shape based approaches are classified. One table gives the list of datasets used, 2nd one gives 3 categories of papers published on 3D human pose from a single monocular image (with datasets used, highlight and metric used). The 3rd table gives the same for 3D human pose estimation from multi-view images. Two more tables gives the same for 3D human pose estimation from a sequence of monocular images and multi-view images. Same is done for multi-person 3D pose estimation. Lot of tables provide performance of the methods (and errors in estimation) on benchmark datasets. The paper concludes that generalization to scenarios in the wild remains extremely challenging. Although few metrics for evaluation studies are discussed, not a single loss/objective function is discussed.

There are a few other papers worth mentioning, which may not provide a thorough survey (or review) encompassing many learning methods used for several CV applications, but concentrate on specific modalities of training or over exclusive task(s), or even on shallow models. For example, [21] discusses imbalanced loss functions on DL models for highly-imbalanced medical image segmentation; [78] discusses issues on de-blurring; [79] gives a thorough and valuable review on 3D reconstruction from stereo views; [80] discusses traditional feature and decision fusion approaches used for shallow models in the field of Pattern Recognition, and [1] provides discussions on evolutionary computation on CV and image analysis.

2.1. Selected Survey papers with focus on loss functions

The paper by Y Tian et. al [37] gives a broad review of the recent progress with loss functions used in deep learning for computer vision tasks. The loss functions for 3 main CV tasks, i.e., object detection, face recognition, and image segmentation are discussed. Specific problems such as imbalanced data, uncertain distribution of the predicted bounding boxes, varied overlapped mode between two bounding boxes and over-fitting are discussed. The survey details the source, derivation, and properties of each loss function. Additionally, advanced challenges about robust losses, generative adversarial networks, noise-tolerant losses, and semantic data augmentation are given. Quite a few loss functions are discussed with diagrams to explain them. They have been categorized based on the 3 application domains targeted in this paper. Many tables are given in the paper, where few provide the names of loss functions for the three applications without explicitly stipulating the expressions, few others list the best SOTA performances, parameters, settings etc. The paper also graphically compares different loss functions and their derivatives. However, it lacks a comprehensive review of many other broader task/application types, learning paradigms employed, and the classifications of loss functions used; all of which have been systematically addressed in our paper. Also, a significant part of the paper focuses on variants of GAN which is not a favored architecture in the modern deep learning era.

Zhang et al. [47] offers an in-depth analysis of recent progress in deep learning methods designed to handle long-tailed data distributions, where a few classes dominate the training data while many others are underrepresented. This imbalance poses a significant challenge to model generalization, particularly for rare "tail" classes. The authors organize existing approaches into three primary categories: class re-balancing (modifying sample distributions or loss functions), information augmentation (such as leveraging transfer learning or generating synthetic data), and module enhancement (improving feature extractors, classifiers, or training strategies). The survey examines these methods in detail, introduces a new evaluation metric called relative accuracy to better assess the performance across class types, and outlines real-world applications and future research directions in this evolving field including handling multi-label long-tailed data, addressing domain shifts, and developing task-specific solutions for applications like medical imaging, species classification, and autonomous driving. Additionally, the paper presents key summaries in Table I (long-tailed dataset statistics), Table II (methods grouped by strategy), and Table III (loss functions used). The surveyed techniques primarily target core computer vision tasks such as classification, detection, segmentation, and visual relation learning. Although it focuses specifically on tackling long-tailed data distributions through targeted strategies and metrics, whereas our survey offers a broader overview of loss functions used across various computer vision tasks, emphasizing their formulation, applicability, and alignment with learning objectives rather than addressing class imbalance directly.

The paper by G Guo et. al [81] summarizes several contributions in the area of Face Recognition (FR). It reviews major ideas on deep learning methods for face image analysis and face recognition. In addition, it provides a concise overview of studies on specific face recognition problems, such as handling variations in pose, age, illumination, expression, and heterogeneous (Cross-modal) face matching. A summary of databases used for deep face recognition is given as well. Tables are provided for a list of deep learning methods based on a single deep CNN, multi-CNN and architectural variants (auto-encoders. LBPnets, Pyramidal networks, GANs , RBM, hybrid models etc.); FR for pose, age-invariance, makeup, video-based, NIR and SWIR-based, 3D data-based FR etc. A section of the paper talks about loss functions, with two large tables provided – one with a list of loss functions with their formulations/expressions and the other gives performances of these. The list is quite elaborate but not grouped (on the table) over various categories, and a bit dated as of current . A discussion on loss functions is provided based on supervision types, such as: sample-based, set-based and others, which is quite crude.

The survey paper by Chen et. al [82] provides a detailed overview of frameworks of both deep and non-deep trackers. Quantitative and qualitative tracking results of various trackers are presented on 5 benchmark datasets with comparative analysis of their results. The paper also discusses challenging circumstances such as occlusion, illumination, deformation, and motion blur. A taxonomy of traditional discriminative trackers is presented. Equations of various traditional filters discussed first, followed by that for CNN-based trackers. Few loss functions are also discussed. Tables are provided for – annotated sequence attributes for tracking benchmarks; list of popular benchmarks; performance comparison of on various datasets; trackers categorized by different motion models; and an exhaustive summarization table (running over 6 pages) for trackers with different attributes - by name, backbone, Dataset, training schemes, update schemes used etc. Experimental results illustrated that recently proposed deep trackers achieved superior performance on public tracking datasets in terms of accuracy, robustness, and speed due to the powerful feature extractor, accurate bounding box regressor, discriminative classifier, and CNNs.

The paper by Rosana E J et. al [83] focuses on concepts of high- level priors embedded at the loss function level, while presenting a survey of the current state-of-the-art methods for medical image segmentation. The articles are categorized according to the nature of the prior: the object shape, size, topology, and the inter-regions constraints. The strengths and limitations of current approaches are highlighted, the challenges related to the design and the integration of prior-based losses and the optimization strategies are discussed. One table provides a list of reviewed papers with respect to the category of the prior types: topology, size, shape and inter-regions priors. The last column of the table provides the optimization function (eg ADMM, SGD etc.) used in each paper. Another table gives targeted segmentation objects and datasets used. Various loss functions used for this task are also discussed separately in the paper. This ranges from CE, Dice etc to high-level prior based penalty functions, based on – topology, shape, inter-regions, adjacency etc. Formulations/expressions of the loss functions are provided and the way this is related to the context and target objects are also discussed. The paper further characterized these methods based on the type of features they optimized, the architecture they use, optimization strategy and the anatomical object that they target. The challenges involved with the design: the fact that the prior must be extracted from soft probability map, the optimization constraints, and the design of the loss with respect to the object of interest, and future trends are also discussed.

A very recent review paper by Edozie et. al [84], 97 pages long, has enormous coverage, with 300 references – covers traditional to modern methods. The paper discusses - Methodology, Feature representation, Datasets, ethical issues etc. Only few (4-5) expressions for loss functions are given, mostly based on– CE, Smoothness etc. This review paper, categorizes current methods into 1-stage and 2-stage frameworks, studying their architectural innovations, detection accuracy, computational speed, deployment readiness, as well as scrutinizes their performance measure. Tables are provided for list of past survey papers, names of OD methods, their strengths, comparative summary of performances over various datasets, key features and drawbacks, evaluation metrics, applications areas (face, pedestrian, vehicle, transportation, surveillance, sports etc.).

The paper by Terven et. al [85] presents a vast and comprehensive review of loss functions and performance metrics in deep learning, highlighting key developments and practical insights across diverse application areas in CV and NLP (Natural Language Processing). The paper is 172 pages long with 350 references. Key contributions of this work include: (1) a unified framework for understanding how losses and metrics align with different learning objectives, (2) an in-depth discussion of multi-loss setups that balance competing goals, and (3) new insights into specialized metrics used to evaluate modern applications like retrieval-augmented generation. There are 12 tables overall, where 2 of them classify few loss functions for CV tasks in 4-5 groups.

Most of the various other survey (and review-based) papers encountered by us ([1, 21, 78, 79, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 42, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163]) do provide a thorough analysis of the architectural models and training paradigms used, as well as datasets used for training & inference. However, they mostly concentrate on a particular application (or related) domain - say Face Recognition (FR), Super Resolution (SR), Object Detection (OD), or segmentation. Some of them even provide a tutorial with a background on CNN literature, such as VAE, LSTM, Transformers, etc.

Using these vast gamut of papers, it becomes difficult for an application developer to now look for, say - what are the various objective (loss) functions used for a particular task or within a learning paradigm? What are the different applications domains or training paradigms, where a particular type of objective function has been used? One may also like to examine the formulation of a loss function, look for similar type (metric structure wise) of objective functions used, and choose a suitable one from a tabular list of the same category. Also, what are the different formulations of loss functions used within a training paradigm, and what all other ones have similar structure (metric-wise) ? None of these papers we have encountered, individually provide answers overall to most of these questions. We attempt to overcome this issue in our paper by grouping or categorizing a bunch of popular loss functions under three different specifications.

2.2. *What does our survey paper provide additionally?*

We attempt to overcome these limitations by providing a more thorough review of almost all application (task) domains, training paradigms, and related loss functions used. The nature and expressions of the loss functions used are mostly missing in most of the review papers published. Our contribution lies in three-fold:

(i) Group the various loss (with names of the cost/objective) functions, describing them briefly with their variations used and list them in tables; Group them over three different categorizations - based on (a) Task/Applications, (b) metric formulation type and (c) Learning paradigms.

(ii) Link the groups as rows, across tables, for obtaining additional information and cross-connectivity over different levels of categorization. This helps a reader to easily identify the set of objective/cost functions used for a particular application, via-a-vis learning paradigm, and obtain their variants too.

(iii) Tiny ablation studies to compare the performance of a few significant (popular) cost functions on an application domain (segmentation/detection) and continual learning based classification. This highlights one of the intended benefits for any researcher/developer using this paper.

(iv) An associated supplementary document to briefly describe the significant loss functions as tabled in this paper.

Comparing two or more objective functions for training a particular DL architecture applied for a CV task is often done using empirical studies. However, it is often beneficial for a researcher/developer (for DL on CV applications) to know and understand the nature of experiment and formulation of a few other similar cost functions published in literature to achieve reasonable performance, and then identify the best one as per SOTA. Our paper takes one step forward in this direction, encompassing various CV applications and learning paradigms used.

We avoid specifically describing any DL architecture used for CV tasks. This is done to keep the length of the paper within reasonable limits. However, there are few other review papers on DL based architectures for CV tasks, which discuss those from CNN to GAN to LSTM, diffusion transformers etc, and we avoid repetition of the same.

3. Categorization of Loss Functions

In this section, we provide 3 sets of categorizations in the form of separate tables, where one can easily identify the various functional forms (with names) of different cost functions used, either for a particular task and or even for a type of training paradigm used. There will be overlap across tables, as a particular type of cost function has been used over a few applications and over different training methods. Significant (based on popularity and use) ones have been briefly described in a supplementary document.

3.1. CATEGORIZATION (TASK-WISE)

The following Table 1 comprises a list of cost/energy functions grouped based on tasks over which they have been used. Brief details of only the significant ones are provided in the supplementary. The last column of Table 1 gives a cross-reference to Table 2, in which the same are grouped based on metric form. Readers are advised to first study each table in isolation, and then look for information linking the different categories.

Table 1: Task-wise categorization of loss functions

Task	S. No.	Loss Function	Formulation	References	Category*
I : Object Detection	1.	Unit correction cost	$c_{i,j}=\lambda c_{\text{loc}}\left(b_i,b_j\right)+(1-\lambda)c_{\text{cls}}\left(p_i,l_i,l_j\right)$	[164]	E
	2.	Angular Margin Loss	$L_{C2AM}=-\log\left(\frac{e^{s\cdot\cos\left(\theta_i\right)}}{e^{s\cdot\cos\left(\theta_i\right)}+\sum_{j=1,j\neq i}^{C}e^{s\cdot\cos\left(\theta_j+m_{ij}\right)}}\right)$	[165]	B, I
	3.	Ratio-preserving loss	$L_{RP}=\frac{1}{N}\left(u\sum_{i\in R_p\cup R_{fp}}L_{SCE}^{i}+v\sum_{i\in R_{tn}}L_{SCE}^{i}\right)$, $u=\frac{N\cdot\alpha}{N_p+N_{fp}},v=\frac{N\cdot(1-\alpha)}{N_{tn}}$	[166, 86]	E
	4.	Hungarian loss	$L_{Hug}=\sum_{i=1}^{N}\left[L_{cls}(c_i,\hat{c}_{\sigma_i})+\mathbb{I}_{\{c_i\neq\varnothing\}}\cdot L_{loc}(b_i,\hat{b}_{\hat{\sigma}_i})\right]$	[167]	F
	5.	Focal loss	$FL\left(p_t\right)=-\left(1-p_t\right)^{\gamma}log(p_t)$	[168, 169]	B
	6.	Side and Corner Align Loss	$SO=\frac{w_{\min}}{w_{\max}}+\frac{h_{\min}}{h_{\max}}$,	[170]	F
	7.	Polygon-to-Polygon Distance Loss	$L_{P2P}=\alpha L_c+\beta L_{sp}+\gamma L_{P2P}^{\prime}$, Where, L'p2p = Polygon-to-polygon loss $L_{P2P}^{\prime}=d\left(A^4,B^4\right)/\left(S_A+S_B\right)$	[171]	F
II :Medical Image Segmentation	8.	Boundary loss	$L_{BD}=\sum_{\Omega}\phi_G(p)s_{\theta}(p)$	[17, 169, 172]	F
	9.	Hausdorff Function	$L_{HD}=\frac{1}{N}\sum_{c=1}^{c}\sum_{i=1}^{N}\left[\left(s_i^c-g_i^c\right)\circ\left(d_{G_i^c}^2+d_{S_i^c}^2\right)\right]$	[18]	A
	10.	Combo loss	$L_{\text{DiceCE}}=L_{CE}+L_{Dice}$	[19]	C
	11.	Sensitivity-specificity loss	$L_{SS}=w\frac{\sum_{c=1}^{c}\sum_{i=1}^{N}\left(g_i^c-S_i^c\right)^2g_i^c}{\sum_{c=1}^{c}\sum_{i=1}^{N}g_i^c+\epsilon}+(1-w)\frac{\sum_{c=1}^{C}\sum_{i=1}^{N}\left(g_i^c-S_i^c\right)^2\left(1-g_i^c\right)}{\sum_{c=1}^{C}\sum_{i=1}^{N}\left(1-g_i^c\right)+\epsilon}$	[20]	D, E
	12.	Asymmetric similarity loss	$L_{Asym}=\left(\sum_{c=1}^{C}\sum_{i=1}^{N}g_i^cs_i^c\right)/\left(\sum_{c=1}^{C}\sum_{i=1}^{N}g_i^cs_i^c+\frac{\beta^2}{1+\beta^2}\sum_{c=1}^{C}\sum_{i=1}^{N}g_i^c\left(1-s_i^c\right)+\frac{1}{1+\beta^2}\sum_{c=1}^{C}\sum_{i=1}^{N}\left(1-g_i^c\right)s_i^c\right)$	[21]	D, F
	13.	Cross Entropy Loss for Segmentation	$L_{WCE}=-\frac{1}{N}\sum_{c=1}^{c}\sum_{i=1}^{N}w_cg_i^c\log s_i^c$	[173]	B
	14.	Distance map penalized cross entropy loss (DPCE)	$L_{DPCE}=-\frac{1}{N}\sum_{c=1}^{C}\left(1+D^c\right)\circ\sum_{i=1}^{N}g_i^c\log s_i^c$	[174]	B
	15.	Dice loss	$L_{\text{Dice}}=1-\frac{2\sum_{c=1}^{c}\sum_{i=1}^{N}g_i^cs_i^c}{\sum_{c=1}^{C}\sum_{i=1}^{N}g_i^c+\sum_{c=1}^{C}\sum_{i=1}^{N}s_i^c}$	[175, 176]	C
	16.	Adaptive Logit Adjustment Loss	$\mathcal{L}_{ALA}=-\log\frac{e^{s\left(f_{\theta}(x)[y]-\mathcal{A}^{ALA}[y]\right)}}{e^{s\left(f_{\theta}(x)[y]-\mathcal{A}^{ALA}[y]\right)}+\sum_{j\neq y}^{C}e^{sf_{\theta}[j]}}$.	[22]	B
	17.	Cross Prototype Contrast	$\mathcal{L}^{\text{cp}}=\frac{1}{\|I\|}\sum_{i\in I}\mathcal{F}\left(\boldsymbol{v}_i;\boldsymbol{y}_i;\mathcal{P}^{\prime}\right)$	[177]	F
	18.	Cross CAM Contrast	$\mathcal{L}^{\text{cc}}=\frac{1}{\|I\|}\sum_{i\in I}\mathcal{F}\left(\boldsymbol{v}_i;\boldsymbol{y}_i^{\prime};\mathcal{P}\right)$	[177]	F
	19.	Background suppression loss	$\mathcal{L}_{CBS}=-\sum_{k=1}^{K}\sum_{l=1}^{L}y_k\cdot\log\left(1-s_{k,l}^{ob}\right)$	[178]	B
	20.	Implicit cross-view consistency Loss	$\mathcal{L}_{\text{reg}}^{\text{mask}}=\sum_{i\in\mathcal{D}}\sum_{v,u\in\mathcal{V},v\neq u}\sum_{c\in\boldsymbol{y}_i}s\left[\mathcal{T}^{(v)\prime}\left(\hat{\mathbf{M}}_{i,c}^{(v)}\right),\mathcal{T}^{(u)\prime}\left(\hat{\mathbf{M}}_{i,c}^{(u)}\right)\right]$	[179]	H

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Table1 – continued from previous page

Task	S. No.	Loss Function	Formulation	Reference	Category*
III :Semantic Segmentation	21.	TopK Loss	$L_{TopK} = -\frac{1}{N}\sum_{c=1}^{c}\sum_{i\in\mathbf{K}} g_i^c \log s_i^c$	[180]	B
	22.	Active Boundary Loss	$\mathbf{ABL} = \frac{1}{N_b}\sum_i^{N_b} \mathbf{\Lambda}(\mathbf{M}_i)\, \mathbf{CE}\left(\mathbf{D}_i^p, \mathbf{D}_i^g\right)$	[181]	F
	23.	Weighted Cross-Entropy Loss	$\mathbf{WCE}(\mathbf{p}, \hat{\mathbf{p}}) = -(\beta\log(\hat{p}) + (1-p)log(1-\hat{p}))$	[169]	B
	24.	Lovász-Softmax loss	$\mathcal{L} = \frac{1}{\lvert\mathcal{C}\rvert}\sum_{c\epsilon\mathcal{C}} \bar{\Delta}_{J_c}(m(c))$	[182]	E
	25.	Anti-noise regularization Loss	$L_{\text{rec}} = \mathcal{L}\left(\mathcal{S}_\theta\left(\{X_t\}, \left\{\hat{Y}_t\right\}, X_1\right), Y_1\right) + \lambda\mathcal{L}_{\text{smooth}}\left(\hat{Y}_t\right)$ $\mathcal{L}_{\text{smooth}}\left(\widehat{Y}_t\right) = \frac{1}{\lvert\Omega\rvert}\sum_{u\in\Omega} \nabla_x \widehat{Y}_{t,u}^2 + \nabla_y \widehat{Y}_{t,u}^2$	[183]	F
IV :Image Enhancement & Restoration	26.	Geometry Loss	$\mathcal{L}_{\text{geo}} = \frac{1}{\lvert\mathcal{P}\rvert}\sum_{\mathbf{x}_p\in\mathcal{P}} \text{BCE}\left(s(\mathbf{x}_p), s^*(\mathbf{x}_p)\right)$	[184]	B
	27.	Texture Loss	$\mathcal{L}_{\text{tex}} = \frac{1}{\lvert\mathcal{R}\rvert}\sum_{\mathbf{r}\in\mathcal{R}} \left\Vert \hat{C}(\mathbf{r}) - C^*(\mathbf{r})\right\Vert^2$	[184]	A
	28.	Pixel Loss	$\mathcal{L}_{pix1} = \frac{1}{WH}\sum_{x=1}^{W}\sum_{y=1}^{H}\left\lvert I_{s(x,y)} - I_{db(x,y)}\right\rvert$ $\mathcal{L}_{pix2} = \frac{1}{WH}\sum_{x=1}^{W}\sum_{y=1}^{H}\left(I_{s(x,y)} - I_{db(x,y)}\right)^2$	[78]	A
	29.	Perceptual Loss	$\mathcal{L}_{per} = \frac{1}{WHC}\sqrt{\sum_{x=1}^{W}\sum_{y=1}^{H}\sum_{c=1}^{C}\left(\Phi_{(x,y,c)}^{l}(I_s) - \Phi^l(x,y,c)^l(I_{db})\right)^2}$	[78]	A
	30.	SRR loss	$\mathcal{L}_{\text{DA}}(\mathcal{D};\theta) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{2}\Vert y_i - \hat{y}(x_i)\Vert_2^2 + \beta\sum_{j=1}^{K}\text{KL}(\hat{\rho}_j \Vert \rho) + \frac{\lambda}{2}\left(\Vert W\Vert_F^2 + \Vert W'\Vert_F^2\right)$	[185]	B, F
	31.	Reconstruction loss	$\mathcal{L}_{\text{recon}} = \sum_{j=\text{ low,normal}}\sum_{j=\text{ low normal }\lambda_{ij}} \lvert \Vert R_j \circ I_j - S_j\Vert_1$	[186]	A
	32.	Illumination Smoothness Loss	$\mathcal{L}_{i.s} = \sum_{i=[aw,normal}\Vert\nabla I_i \circ \exp(-\lambda_g \nabla R_{\dot{r}})\Vert$	[186]	A
	33.	Invariable reflectance loss	$\mathcal{L}_{\text{ir}} = \Vert R_{\text{ow}} - R_{\text{nornal}}\Vert_1$	[185]	A
	34.	Spatial Consistency Loss	$L_{spa} = \frac{1}{K}\sum_{i=1}^{K}\sum_{j\in\Omega(i)}(\lvert(Y_i - Y_j)\rvert - \lvert(I_i - I_j)\rvert)^2$	[187]	A
	35.	Exposure Control Loss	$L_{exp} = \frac{1}{M}\sum_{k=1}^{M}\lvert Y_k - E\rvert$	[187]	A
	36.	Color Constancy Loss	$L_{col} = \sum_{\forall(p,q)\in\varepsilon}(J^p - J^q)^2, \varepsilon = \{(R,G),(R,B),(G,B)\}$	[187]	A
	37.	Self Feature Preserving Loss	$\mathcal{L}_{SFP}\left(I^L\right) = \frac{1}{W_{i,j}H_{i,j}}\sum_{x=1}^{W_{i,j}}\sum_{y=1}^{H_{i,j}}\left(\phi_{i,j}\left(I^L\right) - \phi_{i,j}\left(G\left(I^L\right)\right)\right)^2$	[188]	A
	38.	Content consistency Loss	$L_c = L_{C\_H} + L_{C\_S}$	[189]	E
	39.	SSIM Loss	$\text{SSIM}(x,y) = \frac{2\mu_x\mu_y + c_1}{\mu_x^2 + \mu_y^2 + c_1}\cdot\frac{2\sigma_{xy} + c_2}{\sigma_x^2 + \sigma_y^2 + c_2}$	[190]	A
	40.	Retinex reconstruction loss	$L_r = \Vert\boldsymbol{I} - (\boldsymbol{R}\cdot S + \boldsymbol{N})\Vert_1 + \Vert S - S_0\Vert_1 + \Vert\boldsymbol{R} - \boldsymbol{I}/S\Vert_1$	[191]	E
	41.	Texture enhancement loss	$L_t = \left\Vert w_x\cdot(\partial_x S)^2\right\Vert_1 + \left\Vert w_y\cdot(\partial_y S)^2\right\Vert_1$	[191]	A
	42.	Noise sstimation loss	$L_n = \Vert\boldsymbol{w}_n\cdot\boldsymbol{N}\Vert_F + \frac{1}{\lambda_n}\left[\left\Vert\boldsymbol{w}_r\cdot(\partial_x\boldsymbol{R})^2\right\Vert_1 + \left\Vert\boldsymbol{w}_r\cdot(\partial_y\boldsymbol{R})^2\right\Vert_1\right]$	[191]	A, F
V :Depth Estimation	43.	Photometric loss	$\mathcal{L}_p = \alpha(1 - \text{SSIM}(I_t, I_{t'})) + (1-\alpha)\Vert I_t - I_{t'}\Vert_1$	[192]	A, E
	44.	Edge-aware depth smoothness loss	$\mathcal{L}_{\text{s}}(D_i) = \lvert\delta_x D_i\rvert e^{-\lvert\delta_x I_i\rvert} + \lvert\delta_y D_i\rvert e^{-\lvert\delta_y I_i\rvert}$	[192]	F
	45.	Berhu loss	$\beta_\delta(g,p) = \begin{cases}\lvert g-p\rvert & \text{for } \lvert g-p\rvert \leq \delta \\ \frac{\lvert(g-p)^2\rvert + \delta^2}{2\delta} & \text{otherwise}\end{cases}$	[193]	A
	46.	Stereo-Based Depth Estimation Loss	$E(D) = \sum_x C(x, d_x) + \sum_x\sum_{y\in N_x} E_s(d_x, d_y)$	[79]	F
	47.	3D Supervision Method Loss	$\mathcal{L} = \frac{1}{N}\sum C(x)H(C(x) - \epsilon)\mathcal{D}\left(\Phi(d_x), \Phi\left(\hat{d}_x\right)\right)$	[194]	D, H

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Table1 – continued from previous page

Task	S. No.	Loss Function	Formulation	Reference	Category*
	48.	Uncertainty-Depth Joint Optimization Loss	$\mathcal{L}_{UD} = \frac{1}{N}\sum\left(e^{-s_i}\left(\hat{x}_i - x_i\right)^2 + 2s_i\right)$	[195]	F
	49.	Scale Invariant Gradient Loss	$\mathcal{L}_{\text{grad}}(\xi) = \sum_{h\in\{1,2,4,8,16\}}\sum_{i,j}\left\Vert g^h[\xi](i,j) - g^h[\hat{\xi}](i,j)\right\Vert^2$ where, $g^h[f](i,j) = \frac{f(i+h,j)-f(i,j)}{\lvert f(i+h,j)\rvert + \lvert f(i,j)\rvert}, \frac{f(i,j+h)-f(i,j)}{\lvert f(i,j+h)\rvert + \lvert f(i,j)\rvert}$	[196]	A
VI: Object Localization/Detection	50.	Smooth L-1 Regressor	$L_{\text{reg}}(t^c, v) = \sum_{i\in\{x,y,w,h\}} \text{SmoothL1}(t_i^c - v_i)$ $\text{SmoothL1}(x) = \begin{cases} 0.5x^2 & \text{if } \lvert x\rvert < 1 \\ \lvert x\rvert - 0.5 & \text{otherwise} \end{cases}$	[53]	B, K
	51.	D-IoU and C-IoU	$L_{\text{DIoU}}(B_p, B_t) = 1 - \text{IoU}(B_p, B_t) + \frac{d^2(B_p, B_t)}{c^2}$ $L_{\text{CIoU}}(B_p, B_t) = L_{\text{DIoU}}(B_p, B_t) + \alpha\cdot v$ where, $v = \frac{4}{\pi^2}\left(\arctan\left(\frac{w_t}{h_t}\right) - \arctan\left(\frac{w_p}{h_p}\right)\right)^2$ $\alpha = \frac{v}{(1-\text{IoU})+v}$	[85]	B, K
	52.	Attention-based fusion loss	$L_{at\_fuse} = \left(\frac{V_{x_c}}{\lVert V_{x_c}\rVert_2} - \frac{V_{\tilde{x}_c}}{\lVert V_{\tilde{x}_c}\rVert_2}\right)^2$	[158]	A, J
	53.	Rank and Sort loss	$\mathcal{L}_{\text{RS}} := \frac{1}{\lvert\mathcal{P}\rvert}\sum_{i\in\mathcal{P}}\left(\ell_{\text{RS}}(i) - \ell^*_{\text{RS}}(i)\right)$ $\ell_{\text{RS}}(i) :=$ $\underbrace{\frac{\text{N}_{\text{FP}}(i)}{\text{rank}(i)}}_{\ell_{\text{R}}(i)\,:\,\text{Current Ranking Error}} + \underbrace{\frac{\sum_{j\in\mathcal{P}}\text{H}(x_{ij})(1-y_j)}{\text{rank}^+(i)}}_{\ell_{\text{S}}(i)\,:\,\text{Current Sorting Error}}$	[197]	G
VII: Reconstruction & Super-Resolution	54.	Image Reconstruction Loss	$\mathcal{L} = \frac{1}{N}\sum_x\left\{\left(\nabla_u^2 d_x\right)^2 + \left(\nabla_v^2 d_x\right)^2\right\}$	[198]	G
	55.	Perceptual Loss	$\mathcal{L}_{\text{per}} = \frac{1}{C_iH_iW_i}\left\Vert\phi_i^{\text{vgg}}(I^{\text{SR}}) - \phi_i^{\text{vgg}}(I^{\text{FIR}})\right\Vert_2^2 + \frac{1}{C_jH_jW_j}\left\Vert\phi_j^{\text{tte}}(I^{\text{SR}}) - T\right\Vert_2^2$	[10]	G
	56.	Charbonnier loss	$\mathcal{L}_{\text{pixel\_Cha}}(\hat{I}, I) = \frac{1}{hwc}\sum_{i,j,k}\sqrt{(\hat{I}_{i,j,k} - I_{i,j,k})^2 + \epsilon^2}$	[56]	G
	57.	D-aware contrastive loss	$\mathcal{L}_{3DNCE} = -\log\frac{\sum_{p\in[1..M]} w_{q,p}\cdot\exp(f_q\cdot f_p/\tau)}{M\cdot\sum_{k\in[1..N]}\exp(f_q\cdot f_k/\tau)}$	[199]	B
	58.	Hubber Loss	$\rho(r) = \begin{cases}\frac{r^2}{2} & \text{if } \lvert r\rvert \le c \\ c\left(\lvert r\rvert - \frac{c}{2}\right) & \text{if } \lvert r\rvert > c\end{cases}$	[200]	K
	59.	Tucky Loss	$\rho(r) = \begin{cases}\frac{c^2}{6}\left[1 - \left(1 - \left(\frac{r}{c}\right)^2\right)^3\right] & \text{if } \lvert r\rvert \le c \\ \frac{c^2}{6} & \text{otherwise}\end{cases}$	[200]	K
	60.	Phuber Loss	$\rho(r) = c^2\left(\sqrt{1 + \left(\frac{r}{c}\right)^2} - 1\right)$	[200]	K
Detection	61.	MVI loss	$\text{MVI}(q_S, q_P) = \frac{(H(q_S,q_P) - I(q_S;q_P))I(q_S;q_P)}{H(q_S,q_P)\sqrt{H(q_S)H(q_P)}}$	[159]	B
	62.	Contrastive Loss	$\mathcal{L}_{\text{contrastive}} = \sum_{i=1}^{P}\left((1-y^i)\cdot\frac{1}{2}\left\Vert f(x_1^i) - f(x_2^i)\right\Vert_2^2 + y^i\cdot\frac{1}{2}\left(\max\left(0, m - \left\Vert f(x_1^i) - f(x_2^i)\right\Vert_2\right)\right)^2\right)$	[37]	I

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Table1 – continued from previous page

Task	S. No.	Loss Function	Formulation	Reference	Category*
$VIII$:Face Recognition/	63.	Triplet Loss	$\mathcal{L}_{\text{triplet}} = \sum_{i=1}^{N} \left(\left\Vert f(x_i^q) - f(x_i^p) \right\Vert_2^2 - \left\Vert f(x_i^q) - f(x_i^n) \right\Vert_2^2 + m \right)$	[37]	A, K
	64.	N-pair Loss	$\mathcal{L}_{\text{N-pair}} = \frac{1}{N} \sum_{i=1}^{N} \log \left(1 + \sum_{j \neq i} \exp \left(f(x_i)^\top f(x_j^n) - f(x_i)^\top f(x_i^p) \right) \right)$	[37]	B
	65.	Normalized $\mathcal{X}^2$ Distance (NCD) Loss	$\mathcal{L}_{\text{ncd}}(h(X), h(Y)) = \frac{1}{2\|b\|\|C\|} \sum_{c=1}^{\|C\|} \sum_{i=1}^{\|b\|} \frac{(h(X^c)[i] - h(Y^c)[i])^2}{h(X^c)[i] + h(Y^c)[i]}$	[161]	B
	66.	ArcFace Loss	$L_{C2AM} = -\log \left(\frac{e^{s \cdot \cos(\theta_i)}}{e^{s \cdot \cos(\theta_i)} + \sum_{j=1, j \neq i}^{C} e^{s \cdot \cos\left(\theta_j + m_{ij}\right)}} \right)$	[165]	B, I
IX :Image Classification	67.	Dual Angular Margin Supervised Contrastive Loss	$L_{\text{DAMC}}(\text{proto}, k^+, k^-)$ $= -\frac{1}{\|\mathcal{P}(x)\|} \sum_{k^+} \log \frac{\exp(\text{p}_{\text{proto}}^+/\tau)}{\exp(\text{p}_{\text{proto}}^+/\tau) + \sum_{k^-} \exp(\text{n}_{\text{proto}}^-/\tau)}$	[201]	I
	68.	Supervised contrastive loss	$\mathcal{L}_{\text{cont}}(g; \mathbf{x}, \tau, A) = -\frac{1}{\|P(\mathbf{x})\|} \sum_{\mathbf{k}_+ \in P(\mathbf{x})} \log \frac{\exp(\mathbf{q}^\top \mathbf{k}_+/\tau)}{\sum_{\mathbf{k}' \in A(\mathbf{x})} \exp(\mathbf{q}^\top \mathbf{k}'/\tau)}$	[202]	I
	69.	BCE With Logits Loss	$\mathcal{L}(x, y) = -\left[y \cdot \log\left(\frac{1}{1+e^{-x}}\right) + (1 - y) \cdot \log\left(1 - \frac{1}{1+e^{-x}}\right) \right]$	[203]	B

***Categories (metric) in Table 2**, as referred in last column of Table 1:- **A:** L_*, SSIM; **B:** *log*, ratio, entropy; **C:** Dice, KL, max; **D:** Bayes Theorem; **E:** λ (Linear Combination); **F:** Combo(gen); **G:** Ranking; **H:** Mask, Region; **I:** Contrastive; **J:** Feature; **K:** MISC.

These loss functions in table 1 (also overlapped in Tables 2 & 3) are briefly discussed in the supplementary doc, referring to the paper which define/ specify the same. Few survey papers reviewed by us also discuss the same.

3.2. CATEGORIZATION (METRIC FORM)

The following Table 2 comprises a list of cost/energy functions grouped based on metric form. This table is cross-referenced from Table 1 (last column) in section 3.1, and also similarly from Table 3 in section 3.3.

Table 2: Metric Form based categorization

Metric Form	S. No.	Loss Function	Formulation	Reference
A: L_***, SSIM**	1.	Hausdorff Function	$L_{HD} = \frac{1}{N} \sum_{c=1}^{c} \sum_{i=1}^{N} \left[(s_i^c - g_i^c) \circ \left(d_{G_i^c}^2 + d_{S_i^\varsigma}^2 \right) \right]$	[18]
	2.	Texture Loss	$\mathcal{L}_{\text{tex}} = \frac{1}{\|\mathcal{R}\|} \sum_{\mathbf{r} \in \mathcal{R}} \left\Vert \hat{C}(\mathbf{r}) - C^*(\mathbf{r}) \right\Vert^2$	[184]
	3.	Pixel Loss	$\mathcal{L}_{pix1} = \frac{1}{WH} \sum_{x=1}^{W} \sum_{y=1}^{H} \left\| I_{s(x,y)} - I_{db(x,y)} \right\|$ $\mathcal{L}_{pix2} = \frac{1}{WH} \sum_{x=1}^{W} \sum_{y=1}^{H} \left(I_{s(x,y)} - I_{db(x,y)} \right)^2$	[78]
	4.	Perceptual Loss	$\mathcal{L}_{per} = \frac{1}{WHC} \sqrt{\sum_{x=1}^{W} \sum_{y=1}^{H} \sum_{c=1}^{C} \left(\Phi_{(x,y,c)}^l (I_s) - \Phi^l(x, y, c)^l (I_{db}) \right)^2}$	[78]
	5.	Reconstruction loss	$\mathcal{L}_{\text{recon}} = \sum_{j=\text{ low,normal}} \sum_{j=\text{ low normal } \lambda_{ij}} \| \left\Vert R_j \circ I_j - S_j \right\Vert_1$	[186]
	6.	Illumination Smoothness Loss	$\mathcal{L}_{i.s} = \sum_{i=[aw,normal} \left\Vert \nabla I_i \circ \exp\left(-\lambda_g \nabla R_{\dot{r}}\right) \right\Vert$	[186]
	7.	Invariable reflectance loss	$\mathcal{L}_{\text{ir}} = \left\Vert R_{\text{ow}} - R_{\text{nornal}} \right\Vert_1$	[185]
	8.	Spatial Consistency Loss	$L_{spa} = \frac{1}{K} \sum_{i=1}^{K} \sum_{j \in \Omega(i)} (\|(Y_i - Y_j)\| - \|(I_i - I_j)\|)^2$	[187]
	9.	Exposure Control Loss	$L_{exp} = \frac{1}{M} \sum_{k=1}^{M} \|Y_k - E\|$	[187]

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Table2 – continued from previous page

Metric Form	S. No.	Loss Function	Formulation	Reference
	10.	Color Constancy Loss	$L_{col}=\sum_{\forall(p,q)\in\varepsilon}\left(J^p-J^q\right)^2, \varepsilon=\{(R,G),(R,B),(G,B)\}$	[187]
	11.	Self Feature Preserving Loss	$\mathcal{L}_{SFP}\left(I^L\right)=\frac{1}{W_{i,j}H_{i,j}}\sum_{x=1}^{W_{i,j}}\sum_{y=1}^{H_{i,j}}\left(\phi_{i,j}\left(I^L\right)-\phi_{i,j}\left(G\left(I^L\right)\right)\right)^2$	[188]
	12.	SSIM Loss	$\operatorname{SSIM}(x,y)=\frac{2\mu_x\mu_y+c_1}{\mu_x^2+\mu_y^2+c_1}\cdot\frac{2\sigma_{xy}+c_2}{\sigma_x^2+\sigma_y^2+c_2}$	[190]
	13.	Texture enhancement loss	$L_t=\left\Vert w_x\cdot\left(\partial_x S\right)^2\right\Vert_1+\left\Vert w_y\cdot\left(\partial_y S\right)^2\right\Vert_1$	[191]
	14.	Noise estimation loss	$L_n=\Vert\boldsymbol{w}_n\cdot\boldsymbol{N}\Vert_F+\frac{1}{\lambda_n}\left[\left\Vert\boldsymbol{w}_r\cdot\left(\partial_x\boldsymbol{R}\right)^2\right\Vert_1+\left\Vert\boldsymbol{w}_r\cdot\left(\partial_y\boldsymbol{R}\right)^2\right\Vert_1\right]$	[191]
	15.	Photometric loss	$\mathcal{L}_p=\alpha\left(1-\operatorname{SSIM}\left(I_t,I_{t'}\right)\right)+(1-\alpha)\left\Vert I_t-I_{t'}\right\Vert_1$	[192]
	16.	Berhu loss	$\beta_\delta(g,p)=\begin{cases}\vert g-p\vert & \text{for } \vert g-p\vert\leq\delta\\ \frac{\vert(g-p)^2\vert+\delta^2}{2\delta} & \text{otherwise}\end{cases}$	[193]
	17.	Separable Flow	$L=\sum_{i=0}^N\lambda^{N-i}\left\Vert\mathbf{f}_{gt}-\mathbf{f}_i\right\Vert_1$	[204]
B: log, ratio, entropy	18.	Binary Cross Entropy Loss	$\mathcal{L}_{\mathrm{BCE}}=-[y\cdot\log(\sigma(x))+(1-y)\cdot\log(1-\sigma(x))]$	[205]
	19.	Cross Entropy Loss	$\mathcal{L}_{\mathrm{CE}}=-\sum_{i=1}^C y_i\cdot\log(p_i)$	[205]
	20.	Weighted Cross Entropy Loss	$WCE(p,\hat{p})=-(\beta\log(\hat{p})+(1-p)log(1-\hat{p}))$	[169]
	21.	Label Smoothing Cross Entropy Loss	$\mathcal{L}=-\sum_{i=1}^K q_i\log p_i$ $p_i=\frac{\exp(z_i)}{\sum_{j=1}^K\exp(z_j)}$ $q_i=\begin{cases}1-\varepsilon & \text{if } i=y\\ \frac{\varepsilon}{K-1} & \text{otherwise}\end{cases}$	[206]
	22.	Angular Margin Loss	$L_{C2AM}=-\log\left(\frac{e^{s\cdot\cos(\theta_i)}}{e^{s\cdot\cos(\theta_i)}+\sum_{j=1,j\neq i}^C e^{s\cdot\cos\left(\theta_j+m_{ij}\right)}}\right)$	[165]
	23.	Focal loss	$FL\left(p_t\right)=-\left(1-p_t\right)^\gamma log(p_t)$	[168]
	24.	Adaptive Logit Adjustment Loss	$\mathcal{L}_{ALA}=-\log\frac{e^{s\left(f_\theta(x)[y]-\mathcal{A}^{ALA}[y]\right)}}{e^{s\left(f_\theta(x)[y]-\mathcal{A}^{ALA}[y]\right)}+\sum_{j\neq y}^C e^{sf_\theta[j]}}.$	[22]
	25.	TopK Loss	$L_{TopK}=-\frac{1}{N}\sum_{c=1}^c\sum_{i\in\mathbf{K}}g_i^c\log s_i^c$	[180]
	26.	SRR loss	$\mathcal{L}_{\mathrm{DA}}(\mathcal{D};\theta)=\frac{1}{N}\sum_{i=1}^N\frac{1}{2}\left\Vert y_i-\hat{y}\left(x_i\right)\right\Vert_2^2+\beta\sum_{j=1}^K\mathrm{KL}\left(\hat{\rho}_j\Vert\rho\right)+\frac{\lambda}{2}\left(\Vert W\Vert_F^2+\Vert W'\Vert_F^2\right)$	[185]
	27.	D-aware contrastive loss	$\mathcal{L}_{3DNCE}=-\log\frac{\sum_{p\in[1..M]}w_{q,p}\cdot\exp\left(f_q\cdot f_p/\tau\right)}{M\cdot\sum_{k\in[1..N]}\exp\left(f_q\cdot f_k/\tau\right)}$	[199]
	28.	Temporal Ensembling Loss	$J_{TE}=\frac{1}{\vert B\vert}\sum_{i\in(B\cap L)}\log\mathbf{z}_i\left[y_i\right]+\frac{\omega(t)}{C\vert B\vert}\sum_{i\in B}\left\Vert\mathbf{z}_i-\tilde{\mathbf{z}}_i\right\Vert_2^2,$	[207]
	29.	MVI loss	$\operatorname{MVI}(q_S,q_P)=\frac{(H(q_S,q_P)-I(q_S;q_P))I(q_S;q_P)}{H(q_S,q_P)\sqrt{H(q_S)H(q_P)}}$	[159]
C: Dice, KL, max	30.	Dice loss	$L_{\text{Dice}}=1-\frac{2\sum_{c=1}^c\sum_{i=1}^N g_i^c s_i^c}{\sum_{c=1}^C\sum_{i=1}^N g_i^c+\sum_{c=1}^C\sum_{i=1}^N s_i^c}$	[208, 175]
	31.	Combo loss	$L_{\text{DiceCE}}=L_{CE}+L_{Dice}$	[19]
	32.	Exponential Logarithmic loss	$L_{ELL}=w_{\text{Dice}}E\left[\left(-\log\left(\text{Dice}_c\right)\right)^{\gamma\text{ Dice}}\right]$ $+w_{CE}E\left[w_c\left(-\log\left(s_i^c\right)\right)^{\gamma CE}\right]$	[209]
	33.	Dice loss with focal loss	$L_{\text{DiceFocal}}=L_{\text{Dice}}+L_{\text{Focal}}$	[172]
	34.	Dice loss with TopK loss	$L_{\text{DiceTopK}}=L_{\text{Dice}}+L_{\text{TopK}}$	[210]
	35.	Generalized Wasserstein Dice loss	$L_{\text{Dice}}=1-\frac{2\vert GnS\vert}{\vert G\vert+\vert S\vert}=\frac{\vert\Delta M\vert}{\vert G\vert+\vert S\vert}$	[211]

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Table2 – continued from previous page

Metric Form	S. No.	Loss Function	Formulation	Reference
	36.	Unbiased guidance loss	$L_{UG} = \sum_i KL\left(\mathbf{v}_{t,i}^{M\to Q} \Vert \mathbf{v}_{t,i}^{m\to Q}\right)$	[212]
D: Bayes Theorem	37.	Angular Margin Loss	$L_{C2AM} = -\log\left(\frac{e^{s\cdot\cos(\theta_i)}}{e^{s\cdot\cos(\theta_i)}+\sum_{j=1,j\neq i}^{C} e^{s\cdot\cos\left(\theta_j+m_{ij}\right)}}\right)$	[165]
	38.	Sensitivity-specificity loss	$L_{SS} = w\frac{\sum_{c=1}^{c}\sum_{i=1}^{N}\left(g_i^c - S_i^c\right)^2 g_i^c}{\sum_{c=1}^{c}\sum_{i=1}^{N} g_i^c+\epsilon} + (1-w)\frac{\sum_{c=1}^{C}\sum_{i=1}^{N}\left(g_i^c-S_i^c\right)^2\left(1-g_i^c\right)}{\sum_{c=1}^{C}\sum_{i=1}^{N}\left(1-g_i^c\right)+\epsilon}$	[20]
	40.	Keypoint detection loss	$m_{u,v}^{(i)} = \frac{e^{w_{u,v}^{(i)}}}{\sum_{j=c^{(i)}}^{c^{(i)}+N}\sum_{k=c^{(i)}}^{c^{(i)}+N} e^{w_{j,k}^{(i)}}}$	[213]
E: λ (Linear Combination)	41.	Unit correction cost	$c_{i,j} = \lambda c_{\text{loc}}\left(b_i,b_j\right) + (1-\lambda)c_{\text{cls}}\left(p_i,l_i,l_j\right)$	[164]
	42.	Ratio-preserving loss	$L_{RP} = \frac{1}{N}\left(u\sum_{i\in R_p\cup R_{fp}} L_{SCE}^i + v\sum_{i\in R_{tn}} L_{SCE}^i\right),$ $u = \frac{N\cdot\alpha}{N_p+N_{fp}}, v = \frac{N\cdot(1-\alpha)}{N_{tn}}$	[166]
	43.	Sensitivity-specificity loss	$L_{SS} = w\frac{\sum_{c=1}^{c}\sum_{i=1}^{N}\left(g_i^c - S_i^c\right)^2 g_i^c}{\sum_{c=1}^{c}\sum_{i=1}^{N} g_i^c+\epsilon} + (1-w)\frac{\sum_{c=1}^{C}\sum_{i=1}^{N}\left(g_i^c-S_i^c\right)^2\left(1-g_i^c\right)}{\sum_{c=1}^{C}\sum_{i=1}^{N}\left(1-g_i^c\right)+\epsilon}$	[20]
	44.	Photometric loss	$\mathcal{L}_p = \alpha\left(1-\mathrm{SSIM}\left(I_t,I_{t'}\right)\right) + (1-\alpha)\lVert I_t - I_{t'}\rVert_1$	[192]
F: Combination(gen)	45.	Side and Corner Align Loss	$SO = \frac{w_{\min}}{w_{\max}} + \frac{h_{\min}}{h_{\max}}$	[170]
	46.	Image Reconstruction Loss in Self-Supervised Methods	$\mathcal{L} = \frac{1}{N}\sum_x \mathcal{D}\left(\Phi\left(I_{ref}\right)(x) - \Phi\left(\tilde{I}_{ref}\right)(x)\right)$	[198, 214]
	47.	Scale-Invariant Gradient Loss	$\mathcal{L} = \sum_{h\in A}\sum_x \left\lVert g_h[D](x) - g_h[\hat{D}](x)\right\rVert_2$	[196]
	48.	Dice loss	$L_{\text{dice}} = 1 - \frac{1}{C}\sum_{k=1}^{C}\frac{2\sum_i^N y_i^k p_i^k}{\sum_i^N y_i^k + p_i^k+\epsilon}$	[215]
	49.	Anti-noise regularization Loss	$L_{\text{rec}} = \mathcal{L}\left(\mathcal{S}_\theta\left(\{X_t\},\left\{\hat{Y}_t\right\},X_1\right),Y_1\right) + \lambda\mathcal{L}_{\text{smooth}}\left(\hat{Y}_t\right)$ $\mathcal{L}_{\text{smooth}}\left(\hat{Y}_t\right) = \frac{1}{\vert\Omega\vert}\sum_{u\in\Omega}\nabla_x\hat{Y}_{t,u}^2 + \nabla_y \hat{Y}_{t,u}^2$	[183]
	50.	Polygon-to-Polygon Distance Loss	$L_{P2P} = \alpha L_c + \beta L_{sp} + \gamma L'_{P2P},$ Where, L'p2p = Polygon-to-polygon loss $L'_{P2P} = d\left(A^4,B^4\right) / \left(S_A+S_B\right)$	[171]
	51.	Asymmetric similarity loss	$L_{Asym} = \left(\sum_{c=1}^{C}\sum_{i=1}^{N} g_i^c s_i^c\right) / \left(\sum_{c=1}^{C}\sum_{i=1}^{N} g_i^c s_i^c + \frac{\beta^2}{1+\beta^2}\sum_{c=1}^{C}\sum_{i=1}^{N} g_i^c\left(1-s_i^c\right) + \frac{1}{1+\beta^2}\sum_{c=1}^{C}\sum_{i=1}^{N}\left(1-g_i^c\right)s_i^c\right)$	[21]
	52.	Cross Prototype Contrast	$\mathcal{L}^{\text{cp}} = \frac{1}{\vert I\vert}\sum_{i\in I}\mathcal{F}\left(\boldsymbol{v}_i;\boldsymbol{y}_i;\mathcal{P}'\right)$	[177]
	53.	Active Boundary Loss	$\mathbf{ABL} = \frac{1}{N_b}\sum_i^{N_b}\mathbf{\Lambda}\left(\mathbf{M}_i\right)\mathbf{CE}\left(\mathbf{D}_i^p,\mathbf{D}_i^g\right)$	[181]
	54.	SRR loss	$\mathcal{L}_{\text{DA}}(\mathcal{D};\theta) = \frac{1}{N}\sum_{i=1}^N \frac{1}{2}\lVert y_i - \hat{y}\left(x_i\right)\rVert_2^2 + \beta\sum_{j=1}^K \mathrm{KL}\left(\hat\rho_j\Vert\rho\right) + \frac{\lambda}{2}\left(\lVert W\rVert_F^2 + \lVert W'\rVert_F^2\right)$	[185]
	55.	Noise estimation loss	$L_n = \lVert\boldsymbol{w}_n\cdot\boldsymbol{N}\rVert_F + \frac{1}{\lambda_n}\left[\left\lVert\boldsymbol{w}_r\cdot\left(\partial_x\boldsymbol{R}\right)^2\right\rVert_1 + \left\lVert\boldsymbol{w}_r\cdot\left(\partial_y\boldsymbol{R}\right)^2\right\rVert_1\right]$	[191]
	56.	Edge-aware depth smoothness loss	$\mathcal{L}_{\text{s}}\left(D_i\right) = \vert\delta_x D_i\vert e^{-\vert\delta_x I_i\vert} + \vert\delta_y D_i\vert e^{-\vert\delta_y I_i\vert}$	[192]
	57.	Stereo-Based Depth Estimation Loss	$E(D) = \sum_x C\left(x,d_x\right) + \sum_x\sum_{y\in N_x} E_s\left(d_x,d_y\right)$	[79]

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Table2 – continued from previous page

Metric Form	S. No.	Loss Function	Formulation	Reference
	58.	Uncertainty-Depth Joint Optimization Loss	$\mathcal{L}_{UD}=\frac{1}{N}\sum\left(e^{-s_i}\left(\hat{x}_i-x_i\right)^2+2s_i\right)$	[195]
	59.	Attention-based fusion loss	$L_{at\_fuse}=\left(\frac{V_{x_c}}{\Vert V_{x_c}\Vert_2}-\frac{V_{\tilde{x}_c}}{\Vert V_{\tilde{x}_c}\Vert_2}\right)^2$	[158]
G: Ranking	60.	Triplet Ranking Loss	$L_{\text{tri}}\left(\mathbf{b}_a,\mathbf{b}_p,\mathbf{b}_n\right)=\frac{1}{2}\max\left(D_h\left(\mathbf{b}_a,\mathbf{b}_p\right)-D_h\left(\mathbf{b}_a,\mathbf{b}_n\right)+m_t,0\right)$ s.t. $\mathbf{b}_a,\mathbf{b}_p,\mathbf{b}_n\in\{+1,-1\}^k$	[216]
H: Mask, Region	61.	Implicit cross-view consistency Loss	$\mathcal{L}_{\text{reg}}^{\text{mask}}=\sum_{i\in\mathcal{D}}\sum_{v,u\in\mathcal{V},v\neq u}\sum_{c\in\mathbf{y}_i}s\left[\mathcal{T}^{(v)\prime}\left(\hat{\mathbf{M}}_{i,c}^{(v)}\right),\mathcal{T}^{(u)\prime}\left(\hat{\mathbf{M}}_{i,c}^{(u)}\right)\right]$	[179]
	67.	D-IoU and C-IoU	$L_{\text{DIoU}}(B_p,B_t)=1-\text{IoU}(B_p,B_t)+\frac{d^2(B_p,B_t)}{c^2}$ $L_{\text{CIoU}}(B_p,B_t)=L_{\text{DIoU}}(B_p,B_t)+\alpha\cdot v$ where, $v=\frac{4}{\pi^2}\left(\arctan\left(\frac{w_t}{h_t}\right)-\arctan\left(\frac{w_p}{h_p}\right)\right)^2$ $\alpha=\frac{v}{(1-\text{IoU})+v}$	[85]
	68.	GIoU	$\mathcal{L}_{\text{GIoU}}=1-\text{GIoU}$ $\text{GIoU}=\text{IoU}-\frac{\lvert C\setminus(A\cup B)\rvert}{\lvert C\rvert}$ $\text{IoU}=\frac{\lvert A\cap B\rvert}{\lvert A\cup B\rvert}$	[217]
I: Contrastive	63.	Dual Angular Margin Supervised Contrastive Loss	$L_{\text{DAMC}}(\text{proto},k^+,k^-)$ $=-\frac{1}{\lvert\mathcal{P}(x)\rvert}\sum_{k^+}\log\frac{\exp(\text{p}_{\text{proto}}^+/\tau)}{\exp(\text{p}_{\text{proto}}^+/\tau)+\sum_{k^-}\exp(\text{n}_{\text{proto}}^-/\tau)}$	[201]
	64.	Contrastive Loss with Memory	$\mathcal{L}_{\mathbf{q},\mathbf{k},\mathcal{Q}}=-\log\left(\frac{\exp(\mathbf{q}^\top\mathbf{k}/\tau)}{\exp(\mathbf{q}^\top\mathbf{k}/\tau)+\sum_{\mathbf{n}\in\mathcal{Q}}\exp(\mathbf{q}^\top\mathbf{n}/\tau)}\right)$	[218]
J: Feature	65.	Orthogonal Projection Loss	$\mathcal{L}_{\text{OPL}}=(1-s)+\gamma*\lvert d\rvert$	[219]
K: MISC	66.	Smooth L-1 Regressor	$L_{\text{reg}}(t^c,v)=\sum_{i\in\{x,y,w,h\}}\text{SmoothL1}(t_i^c-v_i)$ $\text{SmoothL1}(x)=\begin{cases}0.5x^2 & \text{if }\lvert x\rvert<1\\ \lvert x\rvert-0.5 & \text{otherwise}\end{cases}$	[53]
	69.	3D Supervision Method Loss	$\mathcal{L}=\frac{1}{N}\sum C(x)H(C(x)-\epsilon)\mathcal{D}\left(\Phi\left(d_x\right),\Phi\left(\hat{d}_x\right)\right)$	[194]

The rows of Table 2 are cross-referred with from two other tables 1 & 3 which helps readers to link subgroups of loss functions that might be similar across applications, learning paradigms, and metric forms, helping to have a comprehensive overview.

3.3. CATEGORIZATION (LEARNING PARADIGM-WISE)

The following Table 3 comprises a list of cost/energy functions grouped based on different learning paradigm. Brief details of only the significant ones are provided in the supplementary document.

Table 3: Learning paradigm based categorization

Learning	S. No.	Loss Function	Formulation	Reference	Category
Unsupervised	1.	Self Feature Preserving Loss	$\mathcal{L}_{SFP}\left(I^L\right)=\frac{1}{W_{i,j}H_{i,j}}\sum_{x=1}^{W_{i,j}}\sum_{y=1}^{H_{i,j}}\left(\phi_{i,j}\left(I^L\right)-\phi_{i,j}\left(G\left(I^L\right)\right)\right)^2$	[188]	A
	2.	Photometric loss	$\mathcal{L}_p=\alpha\left(1-\operatorname{SSIM}\left(I_t,I_{t'}\right)\right)+(1-\alpha)\left\Vert I_t-I_{t'}\right\Vert_1$	[192]	A
	3.	Edge-aware depth smoothness loss	$\mathcal{L}_{\mathrm{s}}\left(D_i\right)=\left\vert\delta_x D_i\right\vert e^{-\left\vert\delta_x I_i\right\vert}+\left\vert\delta_y D_i\right\vert e^{-\left\vert\delta_y I_i\right\vert}$	[192]	F
Supervised	4.	SSIM Loss	$\operatorname{SSIM}(x,y)=\frac{2\mu_x\mu_y+c_1}{\mu_x^2+\mu_y^2+c_1}\cdot\frac{2\sigma_{xy}+c_2}{\sigma_x^2+\sigma_y^2+c_2}$	[190]	A
	5.	Generalized Wasserstein Dice loss	$L_{\text{Dice}}=1-\frac{2\vert GnS\vert}{\vert G\vert+\vert S\vert}=\frac{\vert\Delta M\vert}{\vert G\vert+\vert S\vert};$	[211]	C
	6.	3D Supervision Method Loss	$\mathcal{L}=\frac{1}{N}\sum C(x)H(C(x)-\epsilon)\mathcal{D}\left(\Phi\left(d_x\right),\Phi\left(\hat{d}_x\right)\right)$	[194]	K
	7.	Triplet Ranking Loss	$L_{\text{tri}}\left(\mathbf{b}_a,\mathbf{b}_p,\mathbf{b}_n\right)=\frac{1}{2}\max\left(D_h\left(\mathbf{b}_a,\mathbf{b}_p\right)-D_h\left(\mathbf{b}_a,\mathbf{b}_n\right)+m_t,0\right)$ s.t. $\mathbf{b}_a,\mathbf{b}_p,\mathbf{b}_n\in\{+1,-1\}^k$	[216]	G
One-shot & Few-shot	8.	Ratio-preserving loss	$L_{RP}=\frac{1}{N}\left(u\sum_{i\in R_p\cup R_{fp}}L_{SCE}^i+v\sum_{i\in R_{tn}}L_{SCE}^i\right),$ $u=\frac{N\cdot\alpha}{N_p+N_{fp}},v=\frac{N\cdot(1-\alpha)}{N_{tn}}$	[166]	B
	9.	Uncertainty Loss	$\mathcal{L}_{\text{uncertainty}}=\frac{1}{2}\sum_{o\in\{x,y,w,h\}}\left(\frac{\vert\mu_o-\hat{\mu}_o\vert}{\sigma_o}+\log\sigma_o\right)$	[220]	B
	10.	Contrastive Proposal Encoding (CPE) Loss	$\mathcal{L}_{\text{CPE}}=\frac{1}{N}\sum_{i=1}^{N}f(u_i)\cdot L_{z_i}$ $L_{z_i}=\frac{-1}{N_{y_i}-1}\sum_{j=1,j\neq i}^{N}\mathbb{I}[y_i=y_j]\cdot\log\left(\frac{\exp(z_i\cdot z_j/\tau)}{\sum_{k\neq i}\mathbb{I}[y_k=y_i]\cdot\exp(z_i\cdot z_k/\tau)}\right)$	[221]	I, B
	11.	InfoNCE	$\mathcal{L}_{\text{InfoNCE}}(\theta)=-\mathbb{E}_X\left[\log\frac{f_\theta(x^+,c)}{\sum_i f_\theta(x_i,c)}\right]$	[222]	I, D
Zero-shot	12.	Spatial Consistency Loss	$L_{spa}=\frac{1}{K}\sum_{i=1}^{K}\sum_{j\in\Omega(i)}\left(\vert(Y_i-Y_j)\vert-\vert(I_i-I_j)\vert\right)^2$	[187]	A
	13.	Exposure Control Loss	$L_{exp}=\frac{1}{M}\sum_{k=1}^{M}\vert Y_k-E\vert$	[187]	A
	14.	Color Constancy Loss	$L_{col}=\sum_{\forall(p,q)\in\varepsilon}\left(J^p-J^q\right)^2,\varepsilon=\{(R,G),(R,B),(G,B)\}$	[187]	A
	15.	Texture enhancement loss	$L_t=\left\Vert w_x\cdot\left(\partial_x S\right)^2\right\Vert_1+\left\Vert w_y\cdot\left(\partial_y S\right)^2\right\Vert_1$	[191]	A
	16.	Noise estimation loss	$L_n=\Vert\boldsymbol{w}_n\cdot\boldsymbol{N}\Vert_F+\frac{1}{\lambda_n}\left[\left\Vert\boldsymbol{w}_r\cdot\left(\partial_x\boldsymbol{R}\right)^2\right\Vert_1+\left\Vert\boldsymbol{w}_r\cdot\left(\partial_y\boldsymbol{R}\right)^2\right\Vert_1\right]$	[191]	A , F
Weakly-supervised	17.	Cross Prototype Contrast	$\mathcal{L}^{\text{cp}}=\frac{1}{\vert I\vert}\sum_{i\in I}\mathcal{F}\left(\boldsymbol{v}_i;\boldsymbol{y}_i;\mathcal{P}'\right)$	[177]	I
	18.	Attention-based fusion loss	$L_{at\_fuse}=\left(\frac{V_{x_c}}{\Vert V_{x_c}\Vert_2}-\frac{V_{\tilde{x}_c}}{\Vert V_{\tilde{x}_c}\Vert_2}\right)^2$	[158]	A, J

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Table3 – continued from previous page

Learning	S. No.	Loss Function	Formulation	Reference	Category
Self-supervised	19.	Keypoint detection loss	$m_{u,v}^{(i)} = \frac{e^{w_{u,v}^{(i)}}}{\sum_{j=c^{(i)}}^{c^{(i)}+N} \sum_{k=c^{(i)}}^{c^{(i)}+N} e^{w_{j,k}^{(i)}}}$	[213]	D
	20.	Implicit cross-view consistency Loss	$\mathcal{L}_{\text{reg}}^{\text{mask}} = \sum_{i\in\mathcal{D}} \sum_{v,u\in\mathcal{V},v\neq u} \sum_{c\in\mathbf{y}_i} s \left[\mathcal{T}^{(v)\prime}\left(\hat{\mathbf{M}}_{i,c}^{(v)}\right), \mathcal{T}^{(u)\prime}\left(\hat{\mathbf{M}}_{i,c}^{(u)}\right)\right]$	[179]	H
Semi-supervised	21.	Unbiased guidance loss	$L_{UG} = \sum_i KL\left(\mathbf{v}_{t,i}^{M\to Q} \Vert \mathbf{v}_{t,i}^{m\to Q}\right)$	[212]	C
Continual Learning	22.	Dual Angular Margin Supervised Contrastive Loss	$L_{\text{DAMC}}(\text{proto}, k^+, k^-) = -\frac{1}{\lvert\mathcal{P}(x)\rvert} \sum_{k^+} \log \frac{\exp(\text{p}_{\text{proto}}^{+}/\tau)}{\exp(\text{p}_{\text{proto}}^{+}/\tau) + \sum_{k^-} \exp(\text{n}_{\text{proto}}^{-}/\tau)}$	[201]	I
	23.	Supervised contrastive loss	$\mathcal{L}_{\text{cont}}(g;\mathbf{x},\tau,A) = -\frac{1}{\lvert P(\mathbf{x})\rvert} \sum_{\mathbf{k}_+\in P(\mathbf{x})} \log \frac{\exp(\mathbf{q}^\top \mathbf{k}_+/\tau)}{\sum_{\mathbf{k}'\in A(\mathbf{x})} \exp(\mathbf{q}^\top \mathbf{k}'/\tau)}$	[202]	I
Meta Learning	24.	Meta Regression Loss	$\min_\omega \mathbb{E}_{\mathcal{T}\sim p(\mathcal{T})(\mathcal{D}^{\text{tr}},\mathcal{D}^{\text{val}})\in\mathcal{T}} \sum_{(x,y)\in\mathcal{D}^{\text{val}}} \left[\left(x^\top g_\omega(\mathcal{D}^{\text{tr}}) - y\right)^2\right]$	[223]	A
	25.	Contrastive Loss for Generic Siamese Network	$\mathcal{L}(E,Y) = \sum_{i=1}^{P} (1-Y) * \mathcal{L}_G\left(E(Z_1,Z_2)_i\right) + Y * \mathcal{L}_I\left(E(Z_1,Z_2)_i\right)$	[224]	I, E

***Categories (metric) in Table 2**, as referred in last column of Table 1:- **A:** L_*, SSIM; **B:** *log*, ratio, entropy; **C:** Dice, KL, max; **D:** Bayes Theorem; **E:** λ (Linear Combination); **F:** Combo(gen); **G:** Ranking; **H:** Mask, Region; **I:** Contrastive; **J:** Feature; **K:** MISC.

In the next section, we provide results of a few ablation studies to illustrate the purpose and usefulness of the categorizing tables and groups of loss functions within them, as a case study for the reader to take advantage of our survey paper. The ablation studies involve experiments on (i) Continual Learning using SAVC [202] and MS-DAMC [201], (ii) Object Detection/Classification with YOLOv8 [225], using benchmark datasets.

4. Ablation studies using various cost components.

4.1. Continual Learning

We investigate the impact of different loss functions used in the two-stage training framework of SAVC [202] and MS-DAMC [201]. Specifically, we substitute the default loss components in both Stage 1 (representation learning) and Stage 2 (classification refinement) with a variety of alternatives. Our goal is to understand how each loss function contributes to continual learning performance across multiple sessions and datasets as the effectiveness of continual learning methods depend not only on their architectural innovations but also on the design of their training objectives. In this section, we systematically evaluate how alternative choices for Stage 1 and Stage 2 loss functions impact performance across three benchmark datasets: CUB200, Mini-ImageNet, and CIFAR100. Our experimental configurations are summarized in Table 4.

Experiments	Stage 1 Loss	Stage 2 Loss
SAVC [202]/MS-DAMC [201]	SupCon	Cross-Entropy
Exp-A	Angular Margin Loss	Cross-Entropy
Exp-B	SupCon	Focal Loss
Exp-C	Cross Prototype Contrast	Top-K Loss
Exp-D	SupCon	Adaptive Logit Adjustment

Table 4: Comparison of Stage 1 and Stage 2 Losses across different experiments for contrastive learning.

In Stage 1, the baseline methods employ Supervised Contrastive Loss (SupCon) [202] or Dual Angular Margin Contrast (DAMC) [201]. We choose the Angular Margin Loss [165] and Cross Prototype Contrast Loss [177] to align with the stage's goal of building a robust and discriminative feature space. As indicated in the tables with cross-references, Angular Margin Loss bridges probabilistic separation (Categories B & D in Table 2) with contrastive margin maximization (Category I), enforcing large inter-class margins while preserving intra-class compactness. Cross Prototype Contrast Loss belongs to the multi-term combination metrics group (Category F), enhancing semantic alignment by anchoring features to class prototypes and maximizing separation from others. Their synergy ensures that the initial feature representations are both well-separated and semantically consistent, providing a strong foundation for downstream tasks.

In Stage 2, TopK Loss [180], Focal Loss [168], and Adaptive Logit Adjustment Loss [22] are employed as targeted replacements for standard cross-entropy, a choice again guided by their placement in Table 2's Category B, which encompasses logarithmic ratio-based variants of cross-entropy. These losses share a common heritage in probabilistic classification but introduce distinct mechanisms for addressing continual learning challenges. TopK Loss filters out trivial samples, focusing updates on the most informative predictions; Focal Loss emphasizes hard and underrepresented examples, counter-acting imbalance across tasks; and Adaptive Logit Adjustment Loss dynamically shifts logits to correct class bias, ensuring balanced decision boundaries between old and new classes. Their combined effect preserves discriminative capacity while mitigating catastrophic forgetting, which is essential when adapting models to evolving data distributions in an incremental learning setting.

Observing the performances given in Tables 5, 6, and 7, Exp-A consistently under-performs, with sharper declines in later sessions. This drop indicates weaker generalization and limited transfer of learned features to new tasks. In contrast, Exp-C (CPC + Top-K) achieves competitive or superior accuracy in the first few sessions across datasets, and, most notably in Tables 6 and 7, retains strong accuracy deep into later sessions. This suggests that CPC's prototype anchoring mechanism strengthens inter-class separation during representation learning, while Top-K loss mitigates noisy updates and enhances robustness during classifier refinement. Exp-D (SupCon + ALA) also provides mid-to-late session gains (Table 5), showing ALA's ability to counter class imbalance, whereas Exp-B (SupCon + Focal) offers some benefits early but degrades more rapidly.

Tables 8 and 9 extend these findings to MS-DAMC method. The baseline remains strong, but Exp-D (DAMC + ALA) frequently matches or approaches the baseline, while Exp-C again proves robust, maintaining high early-session performance with minimal degradation in later ones.

Method	Base	Ses1	Ses2	Ses3	Ses4	Ses5	Ses6	Ses7	Ses8	Ses9	Ses10
SAVC [202]	**81.8**	77.9	74.9	70.2	**69.9**	**67.0**	66.1	**65.3**	**63.8**	**63.1**	**62.5**
Exp-A (Angular Margin + CE)	79.6	74.1	69.2	67.5	65.8	63.4	62.9	60.8	60.3	59.1	60.1
Exp-B (SupCon + Focal)	81.2	76.4	71.9	69.5	68.0	66.5	64.5	63.3	63.3	62.6	61.1
Exp-C (Cross-Proto Contrast + Top-K)	81.0	**78.2**	**75.2**	**70.5**	69.1	66.9	**66.2**	65.0	62.8	62.4	61.7
Exp-D (SupCon + ALA)	80.3	76.8	73.9	68.6	68.5	66.5	65.4	64.7	63.1	62.3	61.8

Table 5: Ablation results on CUB200 for different loss configurations in SAVC [202]. Accuracy (%) is reported for base and each incremental session.

Method	Base	Ses1	Ses2	Ses3	Ses4	Ses5	Ses6	Ses7	Ses8
SAVC [202]	**81.1**	76.4	72.4	**68.9**	**66.5**	**62.9**	**59.9**	**58.4**	57.1
Exp-A (Angular Margin + CE)	77.5	68.4	64.0	62.5	60.6	57.5	56.3	54.7	55.9
Exp-B (SupCon + Focal)	79.7	76.1	71.4	68.1	65.8	62.7	59.1	57.9	56.7
Exp-C (Cross-Proto Contrast + Top-K)	**81.1**	**76.6**	**72.6**	68.8	66.3	62.1	59.3	57.8	**57.2**
Exp-D (SupCon + ALA)	79.9	75.8	71.7	68.1	65.8	61.3	58.2	57.6	56.5

Table 6: Session-wise accuracy (%) on Mini-ImageNet for different loss configurations in SAVC [202].

Method	Base	Ses1	Ses2	Ses3	Ses4	Ses5	Ses6	Ses7	Ses8
SAVC [202]	**78.5**	73.3	69.4	64.9	61.7	**59.3**	**57.1**	55.2	**53.1**
Exp-A (Angular Margin + CE)	77.0	69.8	65.4	63.0	59.4	58.4	54.5	53.8	51.8
Exp-B (SupCon + Focal)	78.2	72.8	67.5	63.8	58.2	57.4	56.1	54.8	52.1
Exp-C (Cross-Proto Contrast + Top-K)	78.0	72.9	**69.6**	**65.2**	**61.9**	59.1	57.0	**55.3**	**53.1**
Exp-D (SupCon + ALA)	78.1	73.0	68.4	64.1	60.5	58.6	56.7	54.9	52.6

Table 7: Session-wise accuracy (%) on CIFAR100 for different loss configurations in SAVC [202].

Method	0	1	2	3	4	5	6	7	8	9	10
MS-DAMC [201]	**83.0**	**79.9**	**77.2**	**72.3**	**71.6**	**69.0**	**68.1**	**67.9**	**65.0**	**65.9**	**65.3**
Exp-A (Angular Margin + CE)	80.6	77.1	75.2	69.5	68.8	67.5	66.4	65.7	61.4	59.6	61.8
Exp-B (SupCon + Focal)	81.2	78.4	76.6	71.5	69.6	68.2	67.3	66.8	64.6	63.2	63.1
Exp-C (Cross-Proto Contrast + Top-K)	82.5	78.9	76.3	71.0	69.6	68.2	67.1	66.4	64.1	63.8	63.9
Exp-D (DAMC + ALA)	**83.0**	78.9	76.6	71.5	70.8	68.7	67.6	67.1	64.5	64.7	64.8

Table 8: Session-wise accuracy (%) on CUB200.

Method	Base	Ses1	Ses2	Ses3	Ses4	Ses5	Ses6	Ses7	Ses8
MS-DAMC [201]	**82.9**	**79.9**	**76.9**	**74.0**	**69.4**	**69.0**	**63.5**	**63.0**	**62.8**
Exp-A (Angular Margin + CE)	80.9	77.8	75.1	72.0	67.4	67.9	61.6	61.7	60.4
Exp-B (SupCon + Focal)	81.5	78.4	75.9	72.9	68.3	68.1	62.5	62.1	61.2
Exp-C (Cross-Proto Contrast + Top-K)	80.8	78.4	77.3	72.5	67.7	68.3	61.9	62.0	61.3
Exp-D (DAMC + ALA)	81.5	79.3	77.1	73.6	68.5	68.9	63.2	62.7	62.1

Table 9: Session-wise accuracy (%) on Mini-ImageNet for different loss configurations in MS-DAMC [201].

Experiments	Loss
YOLO8n-cls [225]	BCEWithLogitsLoss
Exp-A	Cross-Entropy
Exp-B	Weighted Cross-Entropy
Exp-C	Label Smoothing Cross-Entropy
Exp-D	Focal Loss

Table 10: Comparison of Losses across different experiments on Image Classification for different loss configurations in MS-DAMC [201].

Method	Pixel Size	Accuracy (Top-1)	Accuracy (Top-5)
YOLO8n-cls [225]	224	**68.3**	**87.9**
Exp-A (Cross-Entropy)	224	68.1	87.7
Exp-B (Weighted Cross-Entropy)	224	68.23	87.76
Exp-C (Label Smoothing Cross-Entropy)	224	68.19	87.5
Exp-D (Focal Loss)	224	63.9	84.79

Table 11: Ablation results on ImageNet for the different loss configuration for Image Classification.

Experiments	Box Loss	Objectness Loss	Classification Loss
YOLOv8n [225]	CIoU	BCE	BCE
Exp-A	GIoU	BCE	BCE
Exp-B	CIoU	Focal Loss	BCE
Exp-C	CIoU	BCE	Cross Entropy Loss
Exp-D	CIoU	Dice Loss	Focal Loss

Table 12: Comparison of different Losses across different experiments for Object Detection.

Experiments	Image Size	mAP
YOLOv8n [225]	640	**37.1**
Exp-A (GIoU+BCE+BCE)	640	36.31
Exp-B (CIoU+Focal Loss+BCE)	640	35.89
Exp-C (CIoU+BCE+Cross Entropy Loss)	640	36.93
Exp-D (CIoU+Dice Loss+Focal Loss)	640	35.7

Table 13: Ablation results on COCO for the different loss configuration for Object Detection.

Across all tested datasets, SAVC and MS-DAMC baselines maintain strong performance. Our findings emphasize that the choice of loss functions at both training stages substantially impacts continual learning effectiveness. While traditional losses like SupCon and Cross-Entropy offer strong baselines, combinations like CPC + Top-K or DAMC + ALA can offer meaningful improvements in certain regimes. This suggests that carefully aligning the inductive biases of loss functions with the evolving nature of incremental data is crucial for scalable and resilient continual learning.

4.2. Image Classification

In this study, we examine the influence of various loss functions on the classification performance of the YOLOv8 [225] architecture, specifically utilizing the YOLOv8n-cls variant. The default loss function employed in the YOLOv8 [225] classification pipeline is the Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss), which serves as our baseline. To assess its relative effectiveness, we perform a comparative analysis using alternative loss functions commonly adopted in image classification: Cross-Entropy, Weighted Cross-Entropy, Label Smoothing Cross-Entropy, and Focal Loss (Category B in Table 2). Despite differing formulations, BCEWithLogitsLoss, Cross Entropy, Weighted Cross Entropy, Label Smoothing Cross Entropy, and Focal Loss all share the common objective of improving classification performance by aligning predicted probabilities with true labels. They operate by minimizing the divergence between the model's output and the ground truth, enhancing representation learning through sharper decision boundaries. While BCE treats classes independently and CE-based losses assume inter-class exclusivity, both frameworks aim to learn discriminative features. Variants like weighting, smoothing, and focal modulation refine this process by improving robustness to class imbalance, label noise and hard samples, ultimately supporting more effective optimization and generalization. The set of experiments and their corresponding loss formulations are summarized in Table 10.

The evaluation is conducted on the ImageNet dataset, with all models trained and tested using a fixed input resolution of 224×224 pixels. As reported in Table 11, the baseline configuration using BCEWithLogitsLoss achieves the highest Top-1 and Top-5 accuracies, recording 68.3% and 87.9%, respectively. While both Weighted Cross-Entropy and Label Smoothing Cross-Entropy yield comparable results, Focal Loss leads to a significant drop in performance, particularly in Top-1 accuracy (63.9%), highlighting its limited utility in this balanced classification scenario.

These results in Table 11 suggest that the default loss employed in YOLOv8 [225] is well-suited for classification tasks in large-scale datasets like ImageNet.

4.3. Object Detection

To evaluate the effect of different loss function configurations on object detection performance, we conducted a comprehensive ablation study using the YOLOv8n [225] model on the COCO dataset with a fixed input resolution of 640×640. The baseline configuration of YOLOv8n [225] adopts a combination of Complete IoU (CIoU) for box regression, Binary Cross-Entropy (BCE) for objectness prediction, and BCE for classification loss. This configuration serves as our reference point.

In this study, we systematically varied the three key loss components—box loss, objectness loss, and classification loss—by substituting them with alternative formulations commonly employed in object detection literature. For the box regression loss YOLOv8n [225] employs CIoU loss to optimize bounding box predictions by considering overlap, center distance, and aspect ratio. Our ablation with GIoU (Category K in Table 2) loss retains the core goal of maximizing localization accuracy but emphasizes area alignment over geometric factors. Both aim to improve spatial alignment between predicted and ground-truth boxes. The default BCE loss encourages the model to distinguish object vs. background. Our replacements as Focal Loss and Dice Loss (Category B/D in Table 2) share this goal but address class imbalance differently: focal loss emphasizes hard negatives, while dice loss measures overlap to handle sparse positives. All aim to refine object confidence prediction. YOLOv8n [225] uses BCE loss per class, treating multi-label outputs independently. In contrast, Cross Entropy and Focal Loss (Category B in Table 2) assume single-label classification and emphasize confident predictions or hard examples. Despite the differences, all strive to improve class label prediction accuracy within detected boxes. The experimental configurations and their corresponding loss combinations are summarized in Table 12.

Quantitative results are reported in Table 13, where the mean Average Precision (mAP) serves as the primary evaluation metric. The baseline YOLOv8n [225] configuration outperformed all variants, achieving the highest mAP of 37.1. Replacing CIoU with GIoU (Exp-A) or modifying the objectness and classification losses (Exp-B, Exp-C, Exp-D) led to consistent degradation in detection performance. Notably, Exp-B and Exp-D, which employed Focal and Dice losses, recorded the lowest mAP scores (35.89 and 35.7, respectively), suggesting that these loss functions may not be optimal for the YOLOv8 architecture in this setting.

These findings underscore the effectiveness of YOLOv8's [225] default loss configuration for object detection. While alternative loss functions may offer theoretical advantages under certain conditions (e.g., class imbalance or noisy labels), they appear suboptimal in the current setup and dataset, reinforcing the robustness of the original design choices in YOLOv8 [225].

5. Conclusions

In this specific survey paper, we have attempted to categorize various cost (loss/objective) functions used in different DL models to solve a wide range of tasks in the Computer Vision domain. We believe that this will help practitioners/developers and definitely researchers working in this field of have a look into a particular basket of such cost functions and either deploy them or create newer variants for potential use to obtain better performance for the task. The three different categorizations which we have provided are : Task/Applications wise, Loss metric functions type (by name) and Learning paradigm, which all are not available (completely, as a whole or together) in earlier published survey/review papers. This paper will thus provide adequate information needed for using DL models for solving CV tasks, in future. We also present brief ablation studies with 3-4 cost functions over two different tasks (as case studies), which will pave the way for readers to do the same over other tasks using the lists we have provided in tables. Readers are advised to refer to the accompanying supplementary doc for brief details of most functions given in tables 1-3, to exploit contents of this article.

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Author Contributions

Riya V, Apu S and Samik B have contributed during the final stages of formatting (in latex) and type-setting of the document.

Sonam G was initially involved in administering the individual tasks of a few co-authors – Riya V, Pooja K, S. Jain, Arti K and Meena P who all (along with herself and those in the acknowledgement section of supplementary doc) contributed almost in equal proportions to type-setting various small sub-sections of the document separately.

Sadbhawana T was involved in formatting the major parts of the tables and arranging the references in bibtex, mid-way of the generation of the manuscript. This helped a lot in the progress of the paper.

Riya contributed to the brief descriptions of a few papers from AIR journal and those on Continual learning. Both Apu and Riya contributed to ablation studies.

S. Das administered this entire process from planning to layout, descriptions, identifying relevant survey/review papers and content generation of the manuscript over an elongated period. He was also actively involved in Conceptualization, Investigation, Visualization, and Supervision of the brief summarization of survey/review papers in SOA and monitoring the progress of the manuscript.

Supplementary Document

`https://www.cse.iitm.ac.in/~vplab/lf_survey/supplementary/lf_supplementary.pdf`

Declaration

The authors declare no competing interests.